

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Belagavi-590018, Karnataka



INTERNSHIP REPORT

ON

“DATA SCIENCE”

Submitted in partial fulfillment of requirements for the award of degree

BACHELORS OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by:

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Under the Guidance of

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Department of Artificial Intelligence and Machine Learning



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

SRI SAIRAM COLLEGE OF ENGINEERING

ANEKAL, BENGALURU -562106

ACADEMIC YEAR: 2025-26

SRI SAIRAM COLLEGE OF ENGINEERING

ANEKAL, BENGALURU -562106

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



CERTIFICATE

This is to certify that **KAVYASHREE N** bearing the USN **1SB22AI031** has carried out the **INDUSTRY INTERNSHIP – BINT803B** work entitled “**DATA SCIENCE**” in **TAKE IT SMART(OPC) Pvt. Ltd**, Bengaluru for the partial fulfillment for the award of Bachelor of Engineering in Department of Artificial Intelligence and Machine Learning in Sri Sairam College of Engineering, Bengaluru under Visvesvaraya Technological University, Belagavi during the year 2025-2026.

Signature of the Head of the Department

Dr. Sivaprakash C

Head of the Department

Department of AIML

Signature of the Principal

Dr. B Shadaksharappa

Principal

Name of examiners

Signature with date

1.

2.

SRI SAIRAM COLLEGE OF ENGINEERING

ANEKAL, BENGALURU -562106

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



CERTIFICATE

This is to certify that this internship report is an original work carried out by me under the guidance of my Internship Mentor/Supervisor, **Mr. S. Vanarasan**, and under the supervision of the Industry Internship Supervisor, **Mr. Mallikarjun kumbar**. This report is submitted as a part of the Internship Course of the Undergraduate Programme of Visvesvaraya Technological University, Jnana Sangama, Machhe, Belagavi – 590018.

Date:

Signature of the Student

Internal Guide

Mr. S. Vanarasan

Assistant Professor

Department of AIML

External Guide

Mr. Mallikarjun kumbar

Director

Take It Smart, Bangalore

INTERNSHIP COMPLETION CERTIFICATE

It is certified that **Mr./Ms. Kavyashree N** Department of Artificial Intelligence and Machine Learning bearing USN **1SB22AI031** of **Sri Sairam College of Engineering** carried out his/her internship from **02/02/2026** to **14/05/2026** in this organization. On the basis of his/her regularity, punctuality, interest shown towards learning skills, team participation, work experience and meeting internship objectives, a score of **100** marks out of 100 is awarded.

Remarks, if any: During this period of this internship program with us we found he/she is sincere and hardworking.

Date: 14/05/2026

Signature of Mentor



Name of Mentor : Mr. Mallikarjun_kumbar

Designation : Director

Address : Take It Smart (OPC) Pvt.Ltd, Vijayanagar, Bangalore

Email : Info@takeitsmart.in



Seal of the Organization

ACKNOWLEDGEMENT

I hereby thank the management for providing an opportunity to study in their esteemed institution.

I extend my sincere gratitude to **TAKE IT SMART(OPC) PVT.LTD** and their dedicated trainers for delivering an insightful and enriching experience in **DATA SCIENCE**. The training not only broadened my understanding but also equipped me with skills aligned with current industry standards and expectations.

I deeply grateful to my beloved Founder Chairman, **MJF. Lion Leo Muthu**, my esteemed CEO, **Dr. Sai Prakash Leo Muthu**, and my respected Chief Operating Officer, **Dr. R. Arunkumar**, for their unwavering support and kind encouragement.

I sincerely thank **Dr. B. Shadaksharappa**, Principal, Sri Sairam College of Engineering, Bengaluru, for his constant encouragement and support in nurturing a positive attitude among students.

I express my heartfelt gratitude and sincere thanks to **Dr. Sivaprakash C**, Professor and Head of the Department of AIML, Sri Sairam College of Engineering, Bengaluru, for his valuable guidance and support in facilitating the successful completion of my internship.

I take this opportunity to express my sincere thanks to my internal guide **Mr. S Vanarasan**, Assistant professor, Department of AIML, Sri Sairam College of Engineering, Bengaluru and external guide **Mr. Mallikarjun kumbar**, Director, Take It Smart(OPC) Pvt.Ltd, Bangalore for guiding and encouraging me throughout internship tenure.

I would like to mention my special thanks to all the faculty members of the Department of Artificial Intelligence and Machine Learning, Sri Sairam College of Engineering, Bengaluru, for their valuable support and guidance.

I finally thank my family and friends who have been encouraging me constantly and inspiring me throughout, without whom this internship has never seen the light of the day.

SRI SAIRAM COLLEGE OF ENGINEERING

ANEKAL, BENGALURU - 562106

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



DECLARATION

I, **KAVYASHREE N**, hereby declare that I have successfully completed my internship titled “**Data Science**” at **Take It Smart(OPC) Pvt. Ltd** during the VIII semester of the B.E. (AIML) degree at Sri Sairam College of Engineering, Anekal, Bangalore, under the esteemed guidance of **Mr. Mallikarjun Kumbar**, Director, Take It Smart(OPC) Pvt. Ltd and **Mr. S Vanarasan**, Assistant Professor, Department of Artificial Intelligence and Machine Learning in Sri Sairam College of Engineering, affiliated with Visvesvaraya Technological University. The work carried out during this internship is original and has not been submitted, either in part or in full, for any other degree or award in any university.

KAVYASHREE N (1SB22AI031)

OFFER LETTER



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TAKE IT SMART (OPC) PVT. LTD

INTERNSHIP OFFER LETTER

This is to confirm that **Kavyashree N** From **Sri Sairam College of Engineering** been offered an Internship in **Data Science** at **Take It Smart (OPC) Pvt. Ltd.**

Internship Details

- **Duration:** 4 Month
- **Start Date:** Jan 20th, 2026
- **Location:** Bengaluru

This internship will provide you with continuous technical mentorship and structured learning modules aimed at bridging the gap between academics and industry. You will be guided by our expert trainers and industry professionals, Enabling you to gain hands-on experience, enhance your technical expertise, and strengthen your practical knowledge.

The program is designed to help you build the Right skills and confidence required to succeed in your future professional career.

We look forward to your valuable contribution and active participation.

Best Regards,

Take It Smart (OPC) Pvt. Ltd.

Director



14, SGN Arcade, 1st Floor, 2nd stage, 1st Main Rd, RPC Layout, Hoshalli Extension,
Stage 1, Vijayanagar, Bengaluru, Karnataka 560040

INTERNSHIP CERTIFICATE



TAKE IT SMART (OPC) PVT LTD

"Recognized by the Government of India's Startup India Initiative"

MSME NO : UDYAM-KR-03-0236520

ISO 9001 : 2015

REG NO : 150027

Internship Certificate

CER ID : TIS-2025- MN155816

This certifies that

KAVYASHREE N

Student bearing USN **1SB22AI031** of **SRI SAIRAM COLLEGE OF ENGINEERING** Has Successfully Completed Internship in **Data Science** From **2nd February 2026 to 14th May 2026**

During this period, the intern exhibited keen interest, professionalism, and willingness to learn, fulfilling all assigned responsibilities to our satisfaction.

We commend their performance and wish them all the best for their future career.

We extend our best wishes for her/his continued success in both her/his personal and professional endeavors.



Mallikarjun Kumbar
Director

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ABSTRACT

This report presents a summary of the Data Science internship completed at Take It Smart (OPC) Pvt. Ltd., where practical exposure and technical training were gained in the fields of Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), and Generative AI. The internship focused on developing industry-oriented analytical and application development skills through hands-on projects, modern frameworks, and real-time implementation. During the internship, knowledge was gained in Python programming, SQL, Exploratory Data Analysis (EDA), MySQL Workbench, Power BI, Streamlit, FastAPI, API integration, Machine Learning, Deep Learning, Retrieval-Augmented Generation (RAG), transformer-based models, and intelligent application development. Multiple projects such as analytical dashboards, recommendation systems, sentiment analysis applications, predictive analytics systems, NLP-based applications, and AI-powered solutions were developed, which enhanced technical understanding in data preprocessing, visualization, model development, deployment, and intelligent automation. The major project undertaken during the internship was the development of a Smart Attendance System using Artificial Intelligence and Computer Vision techniques for automated attendance management through real-time face recognition. The system was developed using Streamlit, OpenCV, MTCNN, FaceNet embeddings, MySQL, Plotly, and Python libraries, and included functionalities such as face detection, facial embedding generation, cosine similarity-based face matching, attendance buffering, duplicate prevention, automated attendance marking, analytical dashboards, CSV export, and email-based attendance alerts. The project demonstrated the practical application of Artificial Intelligence, Deep Learning, Computer Vision, database management, and data analytics in solving real-world educational automation problems. Overall, the internship significantly improved programming knowledge, analytical thinking, problem-solving ability, and practical implementation skills while providing valuable industry exposure and bridging the gap between academic learning and real-world technological applications.

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CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

In today's digital world, organizations and educational institutions are increasingly adopting intelligent technologies to automate manual processes, improve efficiency, and enhance data management. Artificial Intelligence, Data Science, Machine Learning, and Computer Vision are playing a major role in transforming traditional systems into smart and automated solutions. Educational institutions, in particular, require efficient systems for monitoring student activities, managing records, and reducing manual administrative work. Attendance management is one such area where automation can significantly improve accuracy, reliability, and operational efficiency.

Traditional attendance systems are mostly manual, time-consuming, and prone to errors such as proxy attendance, duplicate entries, and inaccurate record maintenance. Manual attendance management also requires additional effort for storing, analysing, and maintaining attendance records over long periods. To overcome these limitations, intelligent attendance systems based on facial recognition and computer vision techniques are becoming increasingly popular.

The main objective of this project is to develop a Smart Attendance System using Artificial Intelligence and Deep Learning techniques for real-time face recognition and automated attendance management. The system uses computer vision algorithms to detect human faces from live camera input, extract facial features, recognize individuals, and automatically mark attendance in a database. The project demonstrates the practical application of Data Science, Machine Learning, Deep Learning, and Computer Vision technologies in solving real-world automation problems.

The implementation of the Smart Attendance System is carried out using modern technologies and frameworks such as Python, Streamlit, OpenCV, MTCNN, FaceNet, MySQL, Plotly, and FastAPI concepts. Python serves as the primary programming language for application development and AI model integration. OpenCV is used for image processing and real-time camera operations, while MTCNN is used for accurate face detection from video frames. FaceNet is used to generate facial embeddings for face recognition and identity matching. Cosine similarity techniques are applied for comparing facial embeddings and identifying individuals with high accuracy.

The system also incorporates database management using MySQL and MySQL Workbench for storing attendance records securely. Streamlit is used to develop an interactive and user-friendly web interface for monitoring attendance, visualizing analytics, and managing system operations. Plotly is used to create analytical dashboards and visual representations of attendance records. Additional functionalities such as duplicate attendance prevention, attendance buffering logic, CSV export, and automated email notifications are also integrated into the system to improve reliability and usability.

During the internship at Take It Smart (OPC) Pvt. Ltd., extensive knowledge and practical experience were gained in Python programming, SQL, Exploratory Data Analysis (EDA), Power BI, Streamlit, API integration, Generative AI, Natural Language Processing (NLP), Machine Learning, Deep Learning, and Retrieval-Augmented Generation (RAG). Multiple projects related to data analytics, intelligent dashboards, recommendation systems, NLP applications, and AI-powered systems were developed during the training period. These projects helped in improving programming skills, analytical thinking, data handling techniques, and real-world application development capabilities.

The Smart Attendance System project demonstrates the integration of multiple Data Science and AI concepts into a single intelligent application. The system not only automates attendance marking but also provides real-time monitoring, analytics, and efficient record management. The use of Deep Learning-based face recognition techniques improves security, reduces manual effort, and increases operational efficiency in educational environments.

From an academic perspective, this project provides practical exposure to Artificial Intelligence, Deep Learning, Computer Vision, database management, and dashboard development. It strengthens technical knowledge in Python programming, real-time AI applications, analytical visualization, and intelligent system design. The project bridges the gap between theoretical concepts and practical implementation by applying modern Data Science technologies to solve real-world problems.

In conclusion, the Smart Attendance System is an intelligent and efficient solution for modern attendance management requirements. The combination of Computer Vision, Deep Learning, Data Analytics, and interactive dashboard technologies makes the project technically advanced, scalable, and practically useful for educational institutions and smart monitoring environments.

CHAPTER 2

ABOUT THE COMPANY

CHAPTER 2

ABOUT THE COMPANY

2.1 Brief History

Take It Smart (OPC) Pvt. Ltd. is an Indian-based engineering and software company headquartered in Bengaluru, Karnataka, India. The company operates as both a product-oriented and service-oriented software organization. The company employs experienced professionals with expertise in handling software, web, and application development projects.

The company started its operations in the year 2018 with a team of two members and was officially registered in the year 2021. Since its establishment, the organization has focused on providing software solutions, technical services, internship programs, and industry-oriented training in modern technologies.

Take It Smart (OPC) Pvt. Ltd. emphasizes practical learning and innovation-driven development. The organization provides opportunities for students and interns to gain real-time exposure through projects, technical mentoring, and software development activities. During the internship, practical knowledge was gained in Python, SQL, Exploratory Data Analysis (EDA), Power BI, MySQL Workbench, Streamlit, FastAPI, API Integration, Machine Learning, Deep Learning, NLP, Generative AI, and Retrieval-Augmented Generation (RAG).

2.2 Major Milestones

Take It Smart (OPC) Pvt. Ltd. has achieved steady growth in the software development and technical training industry since its beginning in 2018. One of the major milestones of the company was its official registration as a private organization in the year 2021.

The company expanded its activities into various domains such as web development, embedded applications, Android application development, IT services, and software solutions. Another important milestone was the introduction of internship and software training programs focused on modern technologies and industry requirements.

The organization also adopted practical project-based learning methodologies where students and interns gained hands-on experience through real-time projects and software application development. Technologies related to Data Science, Artificial Intelligence, Machine Learning,

Deep Learning, and Generative AI were integrated into training and project development activities.

Take It Smart (OPC) Pvt. Ltd. has continuously focused on innovation, technical skill development, and providing industry-oriented learning opportunities for students and professionals.

2.3 Services Offered by the Company

Take It Smart (OPC) Pvt. Ltd. provides various software development, technical training, and IT-related services. The company focuses on delivering software solutions and practical learning opportunities using modern technologies and development frameworks.

The major services offered by the company include:

- Embedded Applications Development
- Web Design and Development
- IT Services
- Android Application Development
- Web-Based Software Solutions
- Mobile-Based Services
- Native Applications
- Hybrid Applications
- Internship and Software Training Programs

The company also provides technical training and internship opportunities in domains such as Python programming, SQL, Data Science, Machine Learning, Deep Learning, Web Development, Artificial Intelligence, and Data Analytics. Students are encouraged to work on practical projects and real-time applications to improve their technical and analytical skills.

During the internship, exposure was gained to technologies such as Streamlit, FastAPI, Power BI, API Integration, NLP, Generative AI, and Machine Learning frameworks through project-based learning and practical implementation.

2.4 Contact Details

Company Name: Take It Smart (OPC) Pvt. Ltd.

Business Address:

14, SGN Arcade, 1st Floor, 2nd Stage, 1st Main Road,
RPC Layout, Hoshalli Extension, Stage 1,
Vijayanagar, Bengaluru, Karnataka – 560040

Mobile: +91-8050104212

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Website: www.takeitsmart.in

2.5 Products of the Company

Take It Smart (OPC) Pvt. Ltd. develops and provides various software solutions, applications, and technical learning platforms focused on modern industry technologies. The company's products and solutions are designed to support software development, web technologies, mobile applications, analytics, and intelligent systems.

The organization develops products related to:

- Web-Based Software Solutions
- Android Applications
- Native Applications
- Hybrid Applications
- Embedded Applications
- Mobile Web Applications
- IT Service Solutions

Overall, the company focuses on providing practical software solutions and technical learning opportunities that help students and professionals gain real-world industry experience and technical knowledge.

CHAPTER 3
TASK PERFORMED

CHAPTER 3

TASK PERFORMED

During my internship at Take It Smart (OPC) Pvt. Ltd., I performed a wide range of technical and analytical tasks related to Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Data Analytics, and Software Development. The internship provided practical exposure to real-time application development, intelligent systems, analytical dashboards, and modern AI-based technologies. Throughout the internship, I actively participated in coding practice, project development, analytical tasks, dashboard creation, and implementation of intelligent applications using various frameworks and tools.

The internship began with Python programming and SQL training, where I performed tasks related to variables, data types, operators, loops, conditional statements, functions, and object-oriented programming concepts. I developed programs involving file handling, exception handling, modular programming, and data manipulation. These tasks helped me strengthen my programming fundamentals and improve logical thinking and problem-solving abilities.

I also worked extensively with relational databases and SQL using MySQL Workbench. I performed tasks such as creating databases and tables, writing SQL queries using SELECT, WHERE, GROUP BY, ORDER BY, HAVING, and aggregate functions, and implementing joins, constraints, views, and normalization concepts. I practiced data insertion, updating, deletion, and retrieval operations, which improved my understanding of database management systems and structured data handling.

A major part of the internship focused on Exploratory Data Analysis (EDA) and data visualization techniques. Using Python libraries such as Pandas and NumPy, I performed data cleaning, preprocessing, filtering, transformation, grouping, and handling missing values. I analysed datasets and generated insights using statistical and analytical methods. Visualization libraries such as Plotly and Matplotlib were used to create interactive charts, graphs, and analytical dashboards for better understanding of data trends and patterns.

I also gained practical exposure to Power BI for business analytics and dashboard development. I performed tasks such as importing datasets, creating KPI cards, generating charts and tables,

applying filters and slicers, and developing interactive dashboards. These tasks helped me understand how raw data can be transformed into meaningful business insights and analytical reports.

During the internship, I worked on multiple Data Science and Machine Learning projects involving recommendation systems, sentiment analysis, predictive analytics, NLP applications, and dashboard development. I learned the workflow of machine learning projects including data preprocessing, feature extraction, model training, evaluation, and prediction. Practical exposure was also gained in Deep Learning concepts and transformer-based NLP applications.

I performed tasks related to Natural Language Processing (NLP) and Generative AI using transformer models and modern AI frameworks. These tasks included text classification, sentiment analysis, next-word prediction, emotion detection, and AI-powered intelligent systems. I also explored Retrieval-Augmented Generation (RAG) concepts and API integration techniques for building interactive AI-based applications.

Another important area of work during the internship was application development using Streamlit and FastAPI. I developed interactive web-based applications, analytical dashboards, recommendation systems, and intelligent monitoring systems with user-friendly interfaces. I learned how to integrate machine learning models into deployable applications and create real-time interactive systems.

The internship also provided practical exposure to dashboard visualization and business analytics concepts through various analytical projects. I worked on projects involving sales dashboards, stock analytics dashboards, restaurant business dashboards, pizza sales dashboards, and customer analysis applications using Dash, Plotly, Streamlit, and Python libraries. These projects improved my understanding of interactive data visualization, KPI monitoring, filtering techniques, and analytical reporting systems.

I additionally worked on intelligent recommendation systems such as book recommendation systems and movie recommendation systems using similarity matrices, API integration, and machine learning concepts. These projects helped me understand recommendation algorithms, content-based filtering techniques, and real-time data fetching methods.

During the internship, I also developed applications related to sentiment analysis, emotion

detection, next-word prediction, and customer churn prediction using Machine Learning and Deep Learning concepts. These projects provided practical exposure to transformer-based models, neural networks, prediction systems, and NLP workflows.

The most significant task performed during the internship was the development of the Smart Attendance System using Artificial Intelligence, Deep Learning, and Computer Vision techniques. The project was developed to automate attendance management using real-time face recognition and intelligent monitoring methods. The system was implemented using Streamlit, OpenCV, MTCNN, FaceNet embeddings, MySQL, Plotly, and Python-based analytical libraries.

In addition to project development, I regularly completed coding assignments, practical exercises, debugging tasks, and technical discussions. I participated in project documentation activities, technical mentoring sessions, and practical implementation reviews that improved my communication skills, analytical thinking, and software development practices.

The internship also provided exposure to API integration, deployment concepts, modular project structuring, and real-world software architecture practices. I learned how to organize projects using separate modules, utility functions, database integration layers, and model management systems for scalable application development.

Another important aspect of the internship was gaining practical understanding of real-world AI workflows and intelligent automation systems. I learned how multiple technologies such as databases, analytical dashboards, AI models, APIs, and visualization tools can be integrated together into a single intelligent application to solve practical problems efficiently.

Overall, the tasks performed during the internship provided extensive practical exposure to Python programming, SQL, Data Analytics, Machine Learning, Deep Learning, NLP, Generative AI, dashboard development, and intelligent application development. The internship helped me apply theoretical concepts to real-world projects, strengthen technical and analytical skills, and gain confidence in developing end-to-end AI-powered systems and data-driven applications.

CHAPTER 4

METHODOLOGY

CHAPTER 4

METHODOLOGY

4.1 System Design

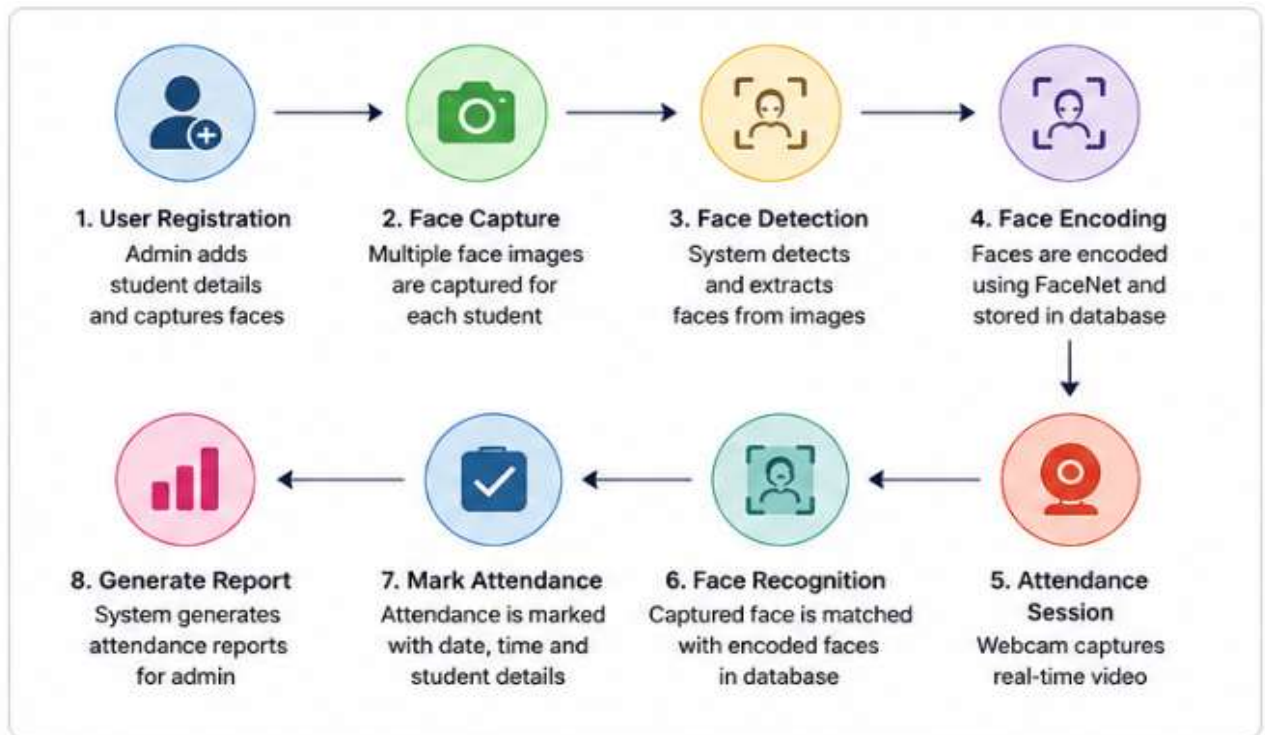


Figure 1: Workflow diagram

The system design of the Smart Attendance System presents the complete architecture and workflow used to automate attendance management using Artificial Intelligence, Computer Vision, and Deep Learning techniques. The project is designed as an integrated intelligent monitoring system that combines real-time face detection, facial recognition, database management, attendance analytics, and automated notification functionalities. The purpose of the system is to reduce manual effort, prevent duplicate attendance, improve accuracy, and provide efficient attendance monitoring through AI-based automation.

The overall system follows a modular layered architecture in which each component performs a specific task and passes the processed output to the next stage. The system integrates multiple technologies such as Streamlit, OpenCV, MTCNN, FaceNet, MySQL, Plotly, and Python libraries to create a complete end-to-end intelligent attendance management solution.

The major layers of the system are:

- Data Collection Layer
- Face Detection Layer
- Face Preprocessing Layer
- Facial Embedding Generation Layer
- Face Recognition and Matching Layer
- Attendance Management Layer
- Database Management Layer
- Analytics and Visualization Layer
- Notification and Export Layer

Each layer is implemented using specialized libraries, frameworks, and tools that work together to provide real-time attendance automation and analytical monitoring.

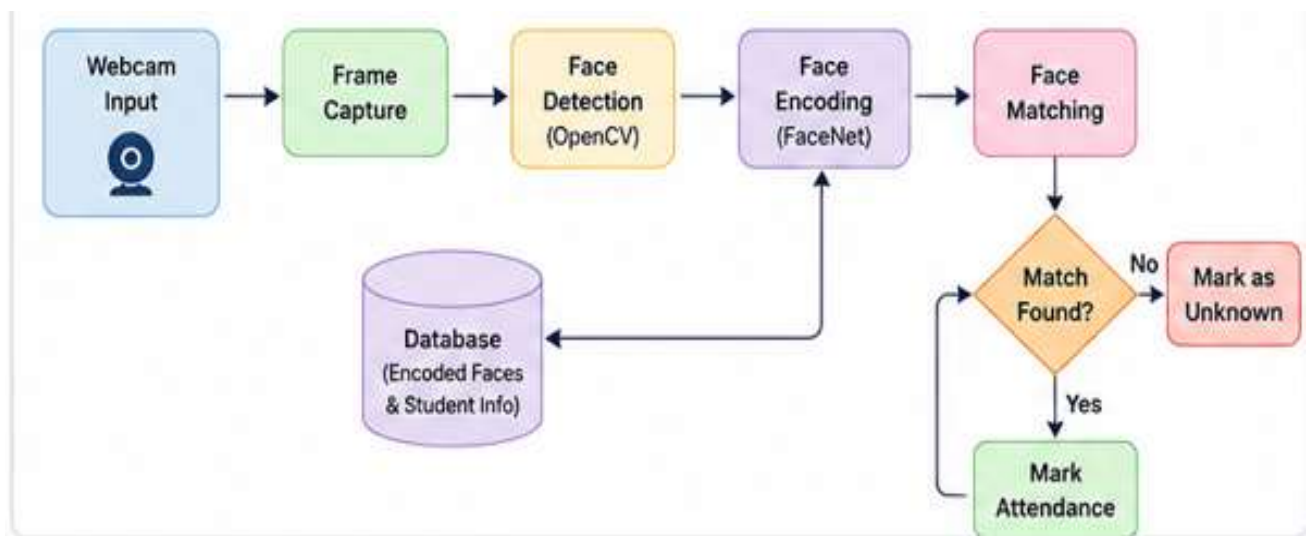


Figure 2: Data flow diagram

Data Collection Layer

The system begins with the collection of face image datasets for registered users. Multiple facial images are captured for each individual using a webcam or camera device. These images are stored inside separate folders based on the person's identity and are used for training and embedding generation.

The dataset preparation process includes:

- Capturing multiple face images for each user

- Organizing images into person-specific folders
- Using real-time camera images for improved accuracy
- Storing images in the dataset directory for embedding generation

The collected facial images serve as the foundation for the face recognition system.

Face Detection Layer

The next stage of the system is the face detection layer, which identifies and extracts human faces from images and live video frames. The project uses the MTCNN (Multi-task Cascaded Convolutional Network) algorithm for accurate face detection.

The face detection layer performs the following tasks:

- Detecting human faces from webcam frames
- Identifying face coordinates
- Eliminating invalid detections
- Ignoring low-quality or very small face regions
- Extracting valid face regions for further processing

OpenCV is used for handling camera operations and image processing tasks, while MTCNN performs robust face detection under different lighting and angle conditions.

Face Preprocessing Layer

After detecting faces, the extracted face regions are passed to the preprocessing layer. This stage prepares facial images in a standardized format suitable for Deep Learning-based embedding generation.

The preprocessing tasks include:

- Converting images from BGR to RGB format
- Resizing face images to 160×160 dimensions
- Normalizing pixel values
- Standardizing image intensity values
- Expanding image dimensions for model compatibility

The preprocessing layer ensures consistency in input data and improves the accuracy of facial

recognition.

Facial Embedding Generation Layer

The Smart Attendance System uses FaceNet for generating facial embeddings. FaceNet is a Deep Learning model that converts facial images into high-dimensional numerical vectors representing unique facial features.

The embedding generation layer performs:

- Feature extraction from facial images
- Conversion of face images into embedding vectors
- Storage of embeddings using Pickle serialization
- Creation of a facial embeddings database for registered users

Each person's dataset images are converted into embeddings, which are later used for similarity comparison during live attendance marking.

Face Recognition and Matching Layer

The face recognition layer is the core intelligence component of the system. During live camera operation, facial embeddings generated from webcam frames are compared with stored embeddings from the database.

The recognition layer performs the following tasks:

- Generating live facial embeddings
- Comparing embeddings using cosine similarity
- Calculating similarity confidence scores
- Identifying the best matching identity
- Applying threshold-based recognition logic
- Marking unknown faces when similarity is low

The system uses cosine similarity for measuring the closeness between embeddings. A predefined threshold is applied to improve recognition accuracy and reduce false matches.

Attendance Management Layer

Once a face is recognized successfully, the attendance management layer automatically records attendance details. The system includes attendance buffering logic to ensure stable recognition before marking attendance.

The attendance management layer performs:

- Duplicate attendance prevention
- Buffer-based stable recognition validation
- Automatic attendance marking
- Real-time attendance updates
- Date and time recording

Attendance records are generated only after multiple successful recognitions, which improves system reliability and prevents accidental marking.

Database Management Layer

The project uses MySQL as the backend database system for securely storing attendance records. Database operations are implemented using the `mysql.connector` library.

The database layer performs:

- Attendance record insertion
- Duplicate attendance checking
- Attendance retrieval
- Date-wise attendance management
- Database connectivity and transaction handling

MySQL Workbench is used for managing database tables and monitoring stored attendance data.

Analytics and Visualization Layer

The Smart Attendance System includes an analytical dashboard developed using Streamlit and Plotly for monitoring attendance records and generating insights.

The analytics layer provides:

- Dashboard overview
- Daily attendance reports
- Monthly attendance analytics
- Person-wise attendance analysis
- KPI cards and attendance statistics
- Interactive charts and visualizations

Plotly is used for creating interactive graphs, pie charts, bar charts, and analytical visualizations that help users understand attendance trends effectively.

Notification and Export Layer

The final layer of the system handles notifications and attendance export functionalities. The project integrates automated email alerts using SMTP protocols.

The notification and export layer performs:

- Automated email notifications after attendance marking
- CSV export functionality for attendance reports
- Real-time success notifications
- Attendance data download support

This layer improves communication, reporting, and usability of the Smart Attendance System.

Overall, the Smart Attendance System integrates Artificial Intelligence, Computer Vision, Deep Learning, database management, and analytical dashboard technologies into a single intelligent platform. The modular system architecture ensures scalability, reliability, real-time monitoring, and efficient attendance automation. The project demonstrates the practical application of modern Data Science and AI technologies in solving real-world educational management problems.

CHAPTER 5

IMPLEMENTATION

CHAPTER 5

IMPLEMENTATION

5.1 Implementation Steps

The implementation of the Smart Attendance System was carried out in a structured and systematic manner using Python, Streamlit, OpenCV, MTCNN, FaceNet, MySQL, Plotly, and related libraries. The development process involved dataset preparation, face detection, preprocessing, embedding generation, face recognition, database integration, attendance automation, and dashboard development. The following steps describe the complete implementation procedure followed in this project.

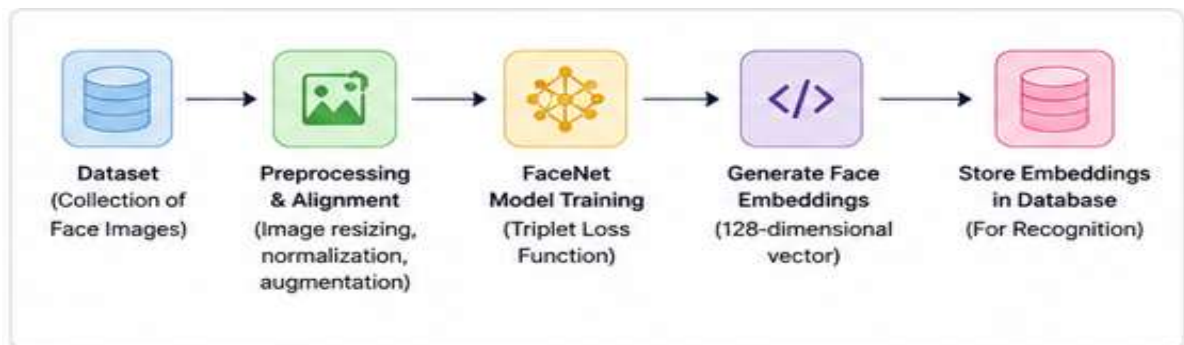


Figure 3: Model training pipeline (FaceNet)

5.1.1 Dataset Collection

The first step of the project was collecting facial image datasets for registered users. Multiple face images of each person were captured using a webcam and stored inside separate folders based on individual names. These images were used for generating facial embeddings and training the recognition system.

The dataset collection process included:

- Capturing multiple face images for each individual
- Storing images in person-specific folders
- Using different face angles and lighting conditions
- Organizing images inside the dataset directory

5.1.2 Face Detection Implementation

The next step involved implementing real-time face detection using the MTCNN (Multi-task Cascaded Convolutional Network) algorithm. OpenCV was used for camera operations and image handling.

The face detection module performed:

- Capturing live video frames from the webcam
- Detecting faces from images and live frames
- Identifying facial coordinates
- Filtering invalid and low-quality detections
- Extracting face regions for preprocessing

This stage ensured accurate face localization before recognition.

5.1.3 Face Preprocessing

After face detection, the extracted face regions were preprocessed before passing them to the Deep Learning model.

The preprocessing tasks included:

- Converting images from BGR to RGB format
- Resizing face images to 160×160 dimensions
- Normalizing pixel intensity values
- Standardizing image data
- Expanding image dimensions for model compatibility

Preprocessing improved consistency and recognition accuracy.

5.1.4 Facial Embedding Generation

The project used FaceNet for generating facial embeddings. The embedding generation process converted facial images into numerical feature vectors representing unique facial characteristics.

The implementation included:

- Loading the FaceNet model
- Passing preprocessed face images to the model
- Generating 512-dimensional embedding vectors
- Creating embeddings for all dataset images
- Storing embeddings using Pickle serialization

The generated embeddings were saved inside the embeddings database file for future recognition tasks.

5.1.5 Attendance Database Creation

A MySQL database was created for storing attendance records securely. Database tables were designed to store attendance information such as:

- Student name
- Attendance status
- Date
- Time

The mysql.connector library was used for database connectivity and query execution. MySQL Workbench was used for database management and monitoring.

5.1.6 Face Recognition and Similarity Matching

The face recognition module compared live webcam embeddings with stored database embeddings using cosine similarity.

The implementation involved:

- Capturing live facial embeddings
- Comparing embeddings using cosine similarity
- Identifying the best matching identity
- Applying similarity threshold logic
- Marking unknown faces when confidence is low

5.1.7 Attendance Automation Logic

After successful recognition, the attendance automation module marked attendance automatically. Buffering logic was implemented to improve recognition stability before attendance marking.

The attendance management process included:

- Stable recognition validation using buffering
- Preventing duplicate attendance entries
- Automatically recording attendance date and time
- Updating attendance records in MySQL database

This module ensured reliable and efficient attendance management.

5.1.8 Email Notification Integration

An automated email notification system was implemented using SMTP protocols. Whenever attendance was marked successfully, email alerts were sent automatically.

The notification module performed:

- Sending attendance confirmation emails
- Integrating Gmail SMTP server
- Handling email authentication securely using environment variables

This feature improved communication and attendance monitoring.

5.1.9 Dashboard Development Using Streamlit and Plotly

An interactive analytical dashboard was developed using Streamlit and Plotly for attendance monitoring and visualization.

The dashboard included:

- Dashboard overview page
- Daily attendance reports
- Monthly attendance analytics

- Person-wise attendance analysis
- KPI cards and attendance statistics
- Attendance tables and CSV export functionality
- Interactive charts and visualizations

5.1.10 Real-Time Camera Integration

The project integrated real-time webcam support using OpenCV. The live camera module continuously processed video frames for face detection and recognition.

The camera implementation included:

- Live webcam capture
- Real-time face recognition
- Bounding box display around detected faces
- Identity label display
- Automated attendance marking during live detection

This feature enabled contactless and automated attendance management.

5.1.11 Testing and Validation

The Smart Attendance System was tested using multiple face images and real-time webcam input. Various scenarios such as different lighting conditions, face angles, and multiple recognition attempts were evaluated.

Testing tasks included:

- Recognition accuracy testing
- Attendance duplicate prevention testing
- Database validation
- Email notification testing
- Dashboard functionality testing
- Real-time camera performance testing

The testing phase ensured proper functioning and stability of the system.

5.2 Hardware and Software Requirements

5.2.1 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 20 GB free disk space or more
- Camera: Webcam or laptop integrated camera
- Display: 1366×768 screen resolution or higher
- Internet Connection: Required for library installation and email services
- Input Devices: Keyboard and Mouse
- System Type: 64-bit computer or laptop

5.2.2 Software Requirements

- Operating System: Windows 10 or Windows 11
- Programming Language: Python 3.x
- Development Environment: VS Code / PyCharm
- Frontend Framework: Streamlit
- Database System: MySQL
- Database Tool: MySQL Workbench
- Documentation Tool: Microsoft Word
- Web Browser: Google Chrome or Microsoft Edge

5.2.3 Python Libraries Used

- OpenCV
- Streamlit
- NumPy, Pandas
- Plotly
- MTCNN
- keras-facenet, pickle
- scikit-learn
- mysql-connector-python
- python-dotenv, smtplib

CHAPTER 6

SCREENSHOTS

CHAPTER 6

SCREENSHOTS

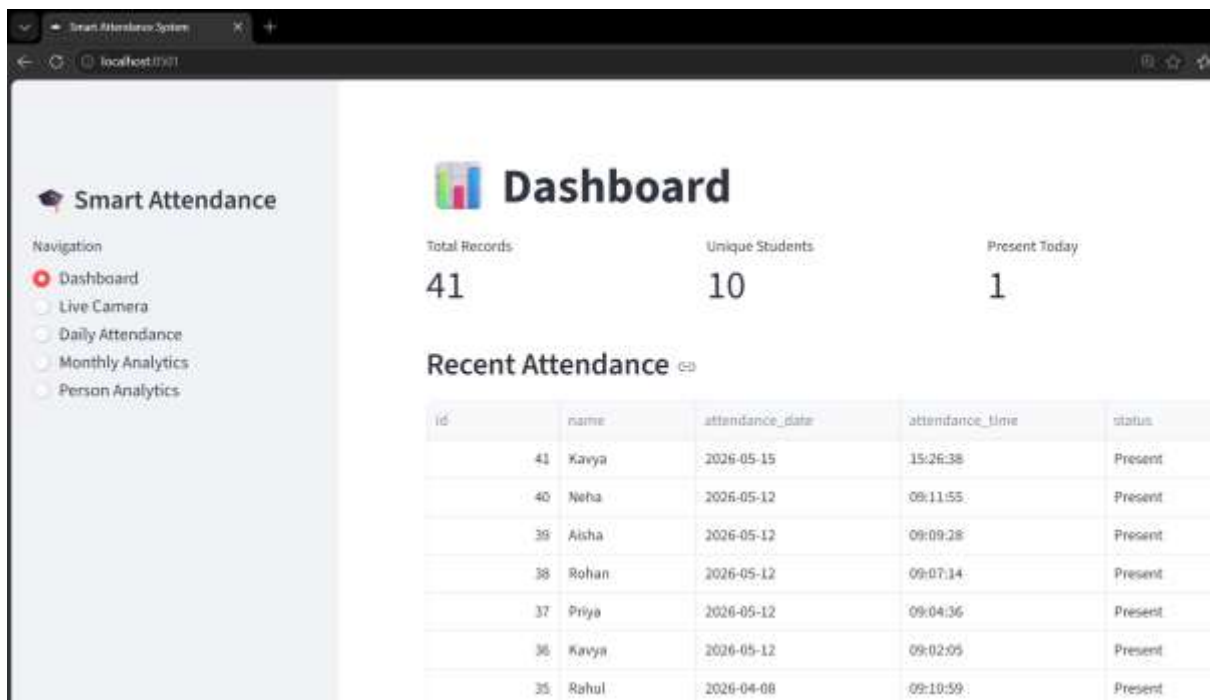


Figure 4: Dashboard

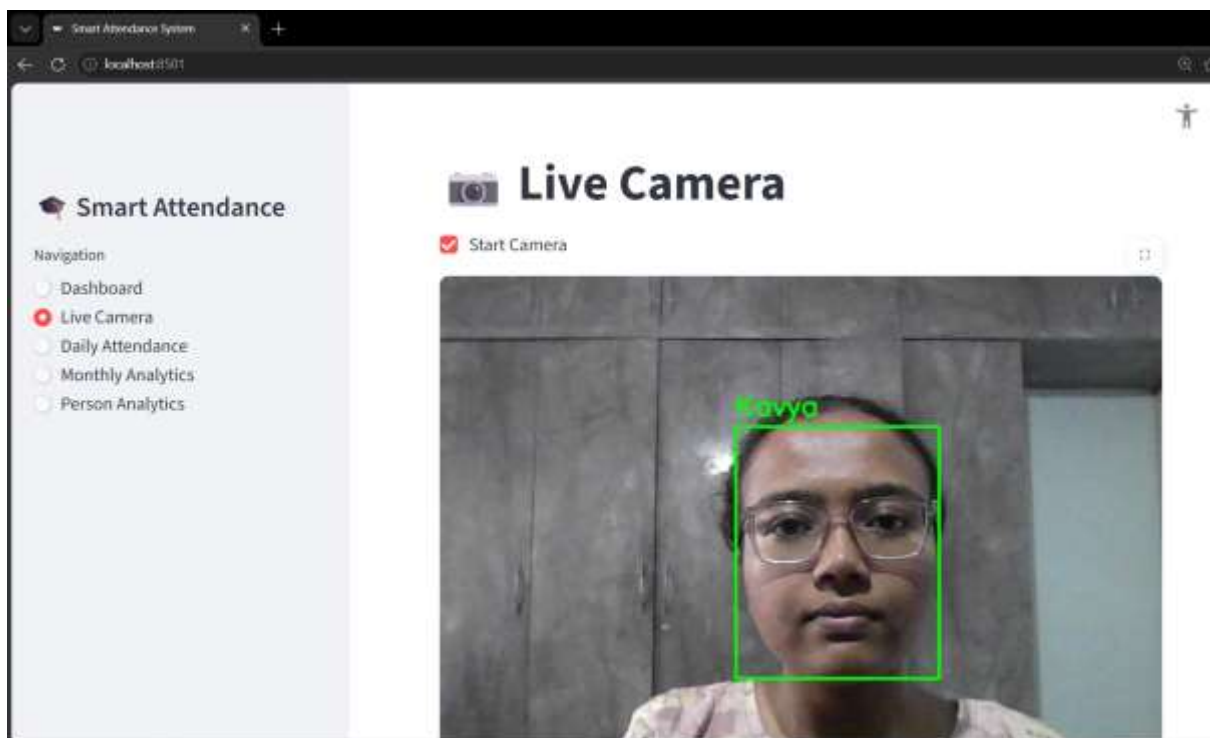


Figure 5: Face detection



Figure 6: Automated Attendance Confirmation Email

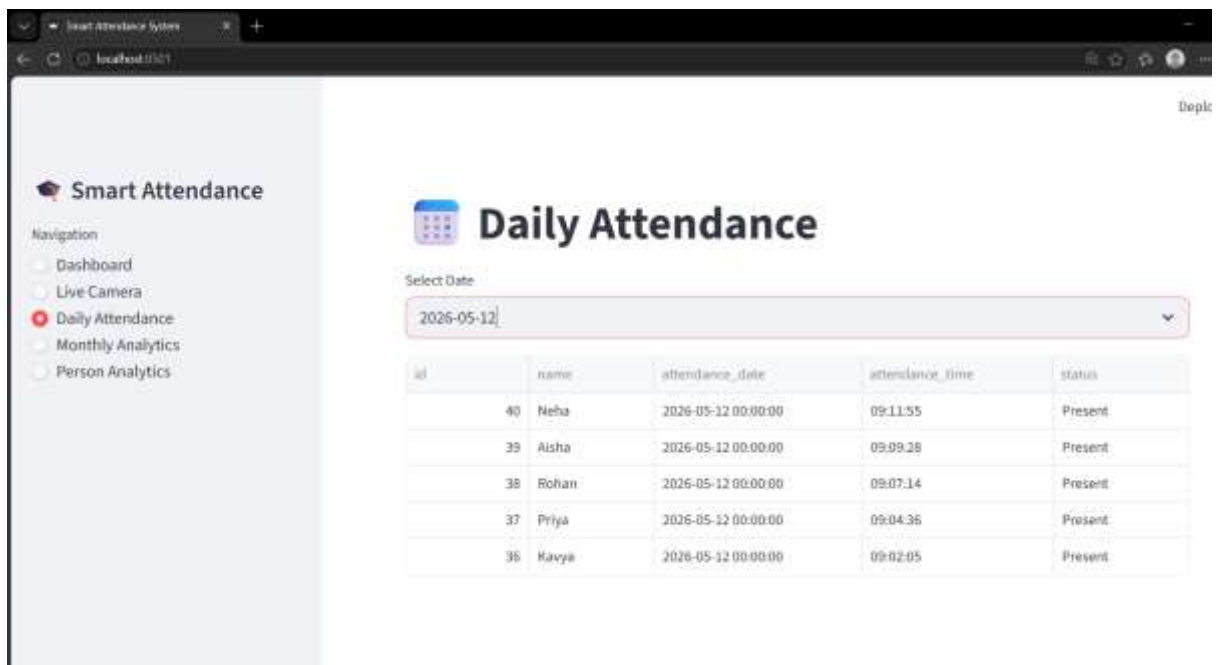


Figure 7: Daily attendance monitoring dashboard.

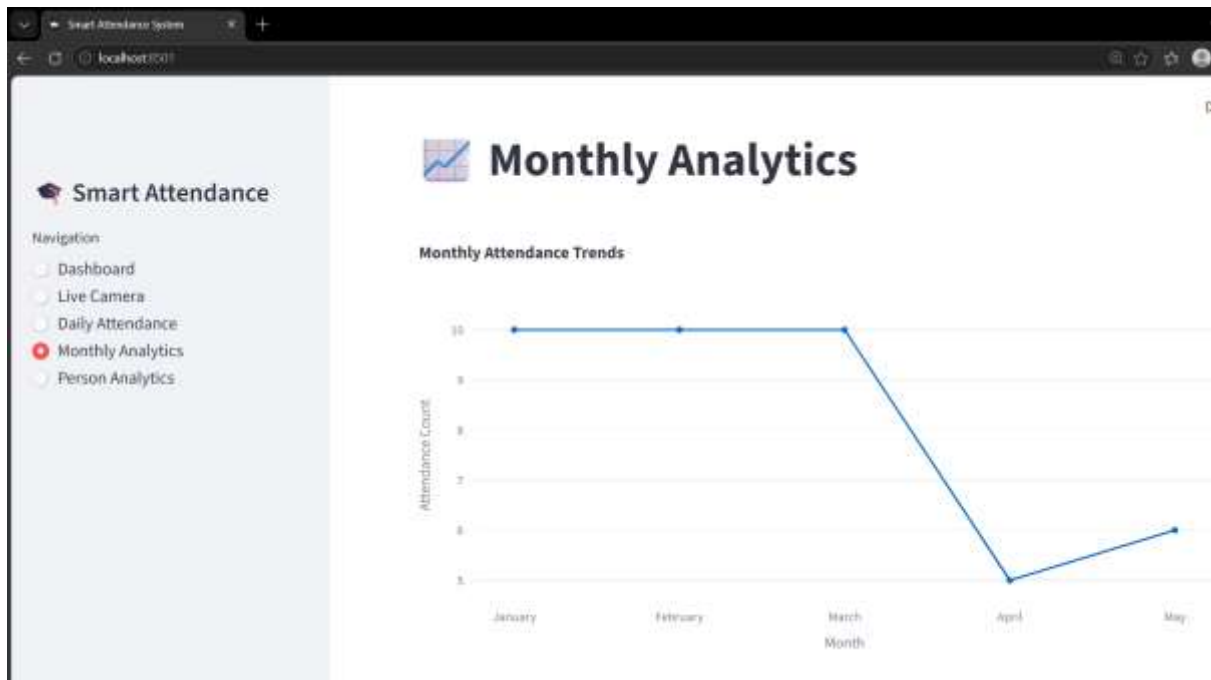


Figure 8: Monthly analytics

Smart Attendance

Navigation

- Dashboard
- Live Camera
- Daily Attendance
- Monthly Analytics
- Person Analytics**

Person Analytics

Select Person

Kavya

Attendance Records - Kavya

id	name	attendance_date	attendance_time	status
1	Kavya	2026-01-05	09:01:12	Present
6	Kavya	2026-01-10	09:02:18	Present
11	Kavya	2026-02-05	09:01:55	Present
21	Kavya	2026-03-05	09:01:20	Present
31	Kavya	2026-04-08	09:01:45	Present
36	Kavya	2026-05-12	09:02:05	Present
41	Kavya	2026-05-15	15:26:38	Present

Figure 9: Person wise attendance analysis

The above screenshots represent the successful implementation and execution of the Smart Attendance System developed using Artificial Intelligence, Computer Vision, Deep Learning, and Data Analytics technologies. The system was designed using Streamlit as the frontend framework along with OpenCV, MTCNN, FaceNet, MySQL, and Plotly for real-time face recognition, attendance management, and analytical visualization. The screenshots demonstrate the different modules and functionalities integrated into the application.

The first screenshot shows the main Dashboard module of the Smart Attendance System. This dashboard acts as the central monitoring interface of the application and displays important attendance statistics such as total attendance records, total unique students, and present attendance count for the current day. The dashboard also displays recent attendance records in tabular format, allowing users to monitor attendance activities efficiently. This module was developed using Streamlit and integrated with the MySQL database for real-time attendance retrieval and display.

The second screenshot represents the Live Camera module, which is one of the core functionalities of the project. In this module, the webcam is activated for real-time face recognition and attendance marking. The system detects the face using the MTCNN face detection algorithm and generates facial embeddings using FaceNet. The recognized face is displayed with a bounding box and identity label around the detected face region. Once the face is recognized successfully, the attendance is automatically marked in the database. This module demonstrates the practical implementation of Computer Vision and Deep Learning techniques in real-time intelligent monitoring systems.

The third screenshot shows the automated email notification generated by the system after successful attendance marking. Whenever attendance is recorded, the application automatically sends a confirmation email to the registered user using SMTP-based email integration. This functionality improves communication and ensures attendance confirmation through automated notifications. The integration of email services demonstrates the implementation of external service connectivity within the application workflow.

The fourth screenshot represents the Daily Attendance module of the application. This module allows users to select a particular date and view all attendance records associated with that date. The attendance details such as student name, attendance date, attendance time, and status are displayed in tabular format. This functionality helps users monitor attendance records efficiently and provides better attendance tracking and management capabilities.

The fifth screenshot shows the Monthly Analytics module developed using Plotly visualization libraries. This analytical dashboard displays attendance trends and monthly attendance statistics through graphical visualizations. The line graph shown in the module represents attendance count trends across different months. The visualization helps in analysing attendance patterns and provides meaningful insights for attendance monitoring and reporting purposes.

The sixth screenshot represents the Person Analytics module of the Smart Attendance System. This module provides person-wise attendance analysis by displaying attendance records of individual users. Users can select a specific person from the dropdown menu and view all associated attendance details including attendance dates, timings, and status. This functionality improves individual attendance monitoring and enables detailed attendance analysis for specific users.

The successful implementation of these modules demonstrates the integration of Artificial Intelligence, Deep Learning, Computer Vision, database management, dashboard visualization, and intelligent automation technologies into a single unified application. The screenshots validate the practical execution of real-time face recognition, attendance automation, analytical reporting, and notification functionalities developed during the internship project. Overall, the Smart Attendance System successfully achieved its objective of providing an intelligent, automated, and efficient attendance management solution using modern AI and Data Science technologies.

CHAPTER 7

REFLECTION NOTES

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REFLECTION NOTES

7.1 Learning from the Internship Program

During the internship at Take It Smart (OPC) Pvt. Ltd., I gained valuable practical knowledge and industry-oriented exposure in Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Data Analytics, and Software Development. The internship helped me strengthen my technical foundation and improve my analytical, logical, and problem-solving abilities through continuous practical implementation and project-based learning.

I learned Python programming concepts including variables, data types, operators, conditional statements, loops, functions, object-oriented programming, file handling, exception handling, and modular programming. Through coding practice, assignments, and project development, I improved my programming knowledge and gained confidence in developing real-world applications using Python.

The internship also provided practical exposure to SQL and database management using MySQL and MySQL Workbench. I learned database creation, table management, query writing, joins, aggregate functions, normalization, constraints, and transaction handling. These concepts helped me understand how structured data is stored, managed, and retrieved efficiently in relational database systems.

I gained hands-on experience in Exploratory Data Analysis (EDA), data preprocessing, and analytical visualization using Python libraries such as Pandas, NumPy, and Plotly. I learned how to clean datasets, handle missing values, perform transformations, analyse trends, and generate meaningful insights through visualizations and analytical reports.

Another important learning outcome of the internship was gaining exposure to Machine Learning and Deep Learning concepts. I learned the workflow of machine learning projects including data preprocessing, feature extraction, model training, evaluation, and prediction. I also understood how AI models can be integrated into real-time applications for intelligent automation and decision-making.

The internship additionally introduced me to Natural Language Processing (NLP), Generative AI, and Retrieval-Augmented Generation (RAG). I learned concepts related to text classification, sentiment analysis, next-word prediction, transformer-based models, and AI-powered intelligent systems. These topics helped me understand modern AI technologies and their practical applications.

I also gained practical knowledge in developing interactive web applications and dashboards using Streamlit and FastAPI. I learned how to create user-friendly interfaces, integrate machine learning models into applications, and develop analytical dashboards for data visualization and monitoring purposes.

One of the most important learning experiences during the internship was the development of the Smart Attendance System using Artificial Intelligence and Computer Vision techniques. Through this project, I learned real-time face detection, facial recognition, image preprocessing, facial embedding generation, cosine similarity matching, attendance automation, database integration, dashboard visualization, and email notification implementation.

The internship also improved my understanding of software architecture, modular project structure, API integration, deployment concepts, and real-world application development practices. I learned how to organize projects into separate modules, utility functions, database layers, and AI model integration systems for scalable development.

Overall, the internship at Take It Smart (OPC) Pvt. Ltd. provided extensive practical exposure to modern technologies such as Data Science, Machine Learning, Deep Learning, NLP, Generative AI, database management, dashboard development, and intelligent application development. It helped bridge the gap between theoretical concepts and practical implementation while preparing me for real-world challenges in the software and AI industry.

7.2 Technical Outcomes

During the internship at Take It Smart (OPC) Pvt. Ltd., I gained significant technical knowledge and practical exposure in programming, analytics, Artificial Intelligence, Machine Learning, Deep Learning, and software development technologies. The internship strengthened my technical foundation by providing practical implementation experience through assignments, coding practice, analytical tasks, and real-time projects.

One of the major technical outcomes of the internship was gaining strong knowledge in Python programming. I learned programming fundamentals such as variables, operators, loops, conditional statements, functions, object-oriented programming concepts, file handling, exception handling, and modular programming. These concepts improved my coding efficiency, logical thinking, and software development understanding.

I also gained practical experience in SQL and database management using MySQL Workbench. I learned database creation, query writing, joins, normalization, aggregate functions, constraints, and transaction handling. This improved my understanding of relational databases and structured data management systems.

The internship enhanced my skills in Exploratory Data Analysis (EDA), data preprocessing, and visualization using Pandas, NumPy, Plotly, and analytical libraries. I learned how to clean datasets, handle missing values, perform data transformations, and create meaningful visualizations for analytical insights and reporting purposes.

Another important technical outcome was gaining exposure to Machine Learning and Deep Learning workflows. I learned concepts related to data preprocessing, model training, prediction systems, feature extraction, and evaluation techniques. I also gained practical experience in integrating AI models into real-time applications and intelligent automation systems.

I gained practical exposure to Natural Language Processing (NLP), transformer-based AI models, and Generative AI applications. I worked on projects related to sentiment analysis, recommendation systems, next-word prediction, emotion detection, and AI-powered intelligent systems using modern AI frameworks.

The internship also improved my skills in developing interactive applications and dashboards using Streamlit and FastAPI. I learned how to build user-friendly interfaces, deploy analytical dashboards, integrate machine learning models, and create intelligent applications with real-time functionalities.

One of the most significant technical outcomes was the successful implementation of the Smart Attendance System project using Computer Vision and Deep Learning technologies. Through this project, I gained practical experience in:

- Face detection using MTCNN
- Facial embedding generation using FaceNet
- Image preprocessing using OpenCV
- Face recognition using cosine similarity
- Real-time webcam integration
- Attendance automation and duplicate prevention
- MySQL database integration
- Dashboard development using Plotly and Streamlit
- Automated email notification systems

This project helped me understand real-time AI application development and intelligent system implementation.

Overall, the internship significantly improved my technical skills in programming, AI, Machine Learning, Deep Learning, analytics, dashboard development, database management, and intelligent application development. The practical exposure gained during the internship enhanced my confidence and prepared me for future opportunities in Data Science, Artificial Intelligence, and software development domains.

7.3 Non-technical Outcomes

During the internship at Take It Smart (OPC) Pvt. Ltd., I also developed several non-technical skills that are important for personal growth and professional success. The internship environment provided opportunities to improve communication skills, teamwork, confidence, analytical thinking, time management, and professional discipline through practical activities and project-based learning.

One of the major non-technical outcomes of the internship was the improvement in communication skills. Regular technical discussions, project explanations, coding presentations, and mentoring sessions helped me improve my ability to explain technical concepts clearly and confidently. These activities enhanced both my verbal and written communication abilities.

The internship also improved my analytical thinking and problem-solving skills. Through coding exercises, debugging tasks, project implementation, and analytical activities, I learned

how to approach problems systematically and identify effective solutions using logical reasoning and practical analysis.

Another important outcome was the development of time management and discipline. Managing multiple assignments, project tasks, coding practice sessions, and internship activities within deadlines helped me understand the importance of planning, consistency, and task prioritization.

The internship additionally improved my teamwork and collaboration abilities. Interacting with mentors and fellow interns during technical discussions and project activities helped me understand the importance of cooperation, knowledge sharing, and effective coordination in a professional environment.

The practical exposure gained during the internship also increased my confidence and professional awareness. Working on real-time projects and implementing AI-based systems helped me become more confident in handling technical tasks and participating in professional discussions. I also gained better understanding of workplace ethics, technical documentation, and industry expectations.

Another important non-technical outcome was the improvement in adaptability and continuous learning. The internship exposed me to multiple modern technologies including AI, Machine Learning, NLP, Generative AI, dashboard development, and software frameworks, which encouraged me to learn new tools and technologies independently.

Overall, the internship contributed significantly to my personal and professional development by improving communication skills, confidence, teamwork, analytical thinking, discipline, adaptability, and professional behaviour. These skills will help me build a strong career in the fields of Data Science, Artificial Intelligence, and software development.

CHAPTER 8

CONCLUSION

CHAPTER 8

CONCLUSION

The internship at Take It Smart (OPC) Pvt. Ltd. was a highly valuable learning experience that provided practical exposure to modern technologies in the fields of Data Science, Artificial Intelligence, Machine Learning, Deep Learning, and Software Development. The internship combined theoretical learning with real-time implementation through coding practice, analytical tasks, dashboard development, and intelligent application development. This helped bridge the gap between academic concepts and industry-level practical applications while improving technical knowledge, analytical thinking, and problem-solving abilities.

During the internship, I gained practical knowledge in Python programming, SQL, Exploratory Data Analysis (EDA), Power BI, MySQL Workbench, Streamlit, FastAPI, API Integration, Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, and Retrieval-Augmented Generation (RAG). I also developed multiple projects related to dashboards, recommendation systems, analytics, NLP applications, and AI-powered systems, which enhanced my understanding of real-world software development and intelligent automation.

The major achievement of the internship was the successful development of the Smart Attendance System using Artificial Intelligence and Computer Vision techniques. The project integrated technologies such as Streamlit, OpenCV, MTCNN, FaceNet, MySQL, Plotly, and Python libraries to automate attendance management through real-time face recognition. The system demonstrated practical implementation of face detection, facial embedding generation, cosine similarity matching, attendance automation, database integration, analytical dashboards, and email notification functionalities.

The internship also improved my understanding of software architecture, database management, dashboard visualization, AI model integration, and real-time application development. Through practical implementation and project-based learning, I gained confidence in handling technical tasks and developing scalable intelligent systems using modern technologies.

In addition to the Smart Attendance System, I worked on several analytical and AI-based projects such as recommendation systems, sentiment analysis applications, customer analytics

systems, dashboards, and predictive applications. These projects helped me understand the complete workflow of data preprocessing, model development, feature extraction, visualization, and deployment. Working on multiple projects also improved my debugging skills, coding efficiency, and ability to integrate different technologies into a single application.

The internship provided practical exposure to real-time dashboard development using Streamlit, Plotly, and Power BI. I learned how to create interactive user interfaces, analytical visualizations, KPI monitoring systems, and real-time reporting dashboards. These experiences improved my understanding of business analytics and data-driven decision-making processes.

I also gained exposure to Natural Language Processing and Generative AI concepts through projects involving sentiment analysis, emotion detection, transformer-based models, and intelligent text processing systems. Practical implementation of these technologies helped me understand how modern AI systems process language data and provide intelligent outputs for real-world applications.

Another important learning outcome of the internship was understanding modular software development and project structuring practices. I learned how to organize applications using separate modules, utility functions, database integration layers, and model management systems for better scalability and maintainability. Exposure to API integration, deployment concepts, and real-time application workflows further improved my understanding of modern software engineering practices.

Apart from technical skills, the internship contributed significantly to my personal and professional development. It improved my communication skills, teamwork, analytical thinking, time management, and professional confidence through technical discussions, coding activities, project documentation, and real-time project implementation.

Overall, the internship at Take It Smart (OPC) Pvt. Ltd. provided valuable industry exposure and strengthened my technical foundation in Data Science and Artificial Intelligence. The knowledge and practical experience gained during the internship will play an important role in my future academic growth and professional career in the fields of AI, Machine Learning, Deep Learning, Data Science, and software development.